

Two mechanisms in the trace–formula analysis of Beyond Endoscopy: the product law and the exact gauge

Abstract

Several estimates on the analytic side of the trace formula are correct *bookkeeping* whose *mechanism* is left outside the frame. We isolate two. The *product law*: a derivative/repulsion statistic is identically a product of distances between configuration points, so “repulsion” is an algebraic identity shared by random matrices and the arithmetic. The *exact gauge*: an oscillatory integrand is a real magnitude times a deterministic phase carrier, so its oscillation is a removable chart artifact and $|\int g| \leq \int |g|$ is lossless.

Working the exact gauge out on the central analytic obstruction of Altuğ’s execution of Beyond Endoscopy for $GL(2)$ — Proposition 5.2 of Part III, the “central issue” of Part II — gives the paper’s four contributions: (1) an *oscillatory cancellation* reducing the uniformity to a single magnitude bound with no oscillation estimate; (2) the identification of the difficulty as a deterministic *emergent clock*, the edge-comb of the fixed cutoff; (3) the consequent *removal of the productivity boundary*, re-read as gradual attrition rather than a wall; and (4) a *candidate automorphic object* in the primary representation, whose four converse-theorem inputs — functional equation, continuation, exhaustion, and loss-ledger recovery — are formalized and machine-checked.

The same two mechanisms organize three further objects: the GUE-level derivative statistics of zeros, the archimedean transfer factor and arithmetic transparency of symmetric-power functoriality, and a detection identity we close on a genuine transfer — on a CM form the pole order of a self-twist Rankin–Selberg L -function, read as the zero-frequency residue of a logarithmic derivative, equals a geometric self-twist relation on Frobenius, both giving the transferred count 1; on a non-CM form both give 0. The uniformity statements are theorems; the detection identity rests on classical Gelbart–Jacquet/Rankin–Selberg theory; the extension to a full symmetric-power functoriality theorem is the subject of active research, with its remaining ingredients stated precisely.

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1 Two mechanisms

Two devices recur in the analytic theory as *answers* while their *explanations* stay outside the frame.

The product law (self-interference). The unitary ensembles predict the local statistics of the zeros of an L -function correctly, but the ensemble contains no mechanism: “eigenvalue repulsion” is postulated. Yet the derivative statistic that encodes repulsion is an identity: for a polynomial $P = \prod_j (X - r_j)$ and a simple root μ ,

$$P'(\mu) = \prod_{r_j \neq \mu} (\mu - r_j), \quad (1)$$

so $|P'(\mu)|$ is the product of distances from μ to the other configuration points. The same product governs the reopening rate of an arithmetic fiber at a zero. Hence a “GUE-level” prediction and its arithmetic counterpart compute one object; they differ only through the configuration fed to (1). Repulsion is the product law, not a force. (This is an algebraic identity; it does not by itself prove that zeros follow GUE.)

The exact gauge (removable oscillation). An oscillatory integral $\int g$ is normally attacked by estimating cancellation. But if $g = |g| e^{i\phi}$ with ϕ a *deterministic* phase carrying no arithmetic content, then the oscillation is a choice of chart and $|\int g| \leq \int |g|$ is lossless. Choosing the gauge that makes the integrand real turns an oscillation estimate into a magnitude bound; over a root-of-unity carrier (the ζ_6 step $2\pi/6$, which closes) the gauged object is a real harmonic.

Sections 2–4 are the three closes: the exact gauge on the archimedean uniformity (§2); the identification of the difficulty as the cutoff’s emergent clock (§3); and the productivity boundary re-read (§4). Sections 5–9 then develop the two mechanisms toward symmetric-power functoriality — the product law and the transfer factor (§5), a carrier/fiber rescaling in which a pole becomes a closed cell (§6), the arithmetic transparency of the transfer (§7), a detection identity closed on a genuine example (§8), and a candidate automorphic object (§9).

2 The exact gauge on the archimedean uniformity

Altuğ [1, 2, 3] executes Beyond Endoscopy for $\mathrm{GL}(2)$ and the standard representation. Its analytic core is a uniformity estimate: after the arithmetic weights $L(1, \chi_D)$ are expanded by an approximate functional equation and the trace variable is Poisson-summed, one must bound, uniformly in (ξ, ν, l, f, X) , the Fourier transform $J_{l,f}(\xi, \nu, X)$ of a product of a fixed cutoff G and a smoothed

orbital transform (Prop. 5.2, Part III); Altuğ calls the C, D -independence of the implied constant “the central issue” [2, App. A]. The transform whose asymptotics drive the appendix is (Part II, Def. A.6)

$$A_{h_a, m}^{\tau, \pm}(\Phi)(x) = \frac{1}{2\pi i} \int_{(\tau)} \tilde{\Phi}(u) c_m^{\pm}(u/2) \Gamma\left(m+1 + \frac{a+u}{2}\right) x^{-u/2} du, \quad \tau > 0, \quad (2)$$

for a singular profile $h_a(x) = |1 - x^2|^{a/2} h_1(x)$ (h_1 smooth, $a > -1$) and Schwartz Φ ; the parameters enter only through $x = \mp C^2 D$ and a unimodular prefactor.

Lemma 1 (the gauge bound). *For every $\tau > 0$, $m \in \mathbb{N}$, and Schwartz Φ ,*

$$|A_{h_a, m}^{\tau, \pm}(\Phi)(x)| \leq B_{\tau, m, a}(\Phi) x^{-\tau/2}, \quad B_{\tau, m, a}(\Phi) := \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \tilde{\Phi}(\tau + iv) c_m^{\pm}\left(\frac{\tau + iv}{2}\right) \Gamma\left(m+1 + \frac{a+\tau}{2} + \frac{iv}{2}\right) \right| dv,$$

with $B_{\tau, m, a}(\Phi) < \infty$ independent of x (hence of C, D).

Proof. On $u = \tau + iv$ one has $|x^{-u/2}| = x^{-\tau/2}$, so the triangle inequality gives the bound and B contains no x . For finiteness: by Stirling $|\Gamma(\sigma + iv/2)| \asymp e^{-\pi|v|/4}$; the phase $i^{1+m+(a+u)/2}$ has magnitude $e^{-\pi v/4}$; and $|\tilde{\Phi}(\tau + iv)| \ll e^{-\pi|v|/2}$. The product decays like $e^{-\pi|v|}$ as $v \rightarrow +\infty$ and $e^{-\pi|v|/2}$ as $v \rightarrow -\infty$ (there the Γ - and phase-magnitudes cancel and $\tilde{\Phi}$ carries the decay); the Γ -poles lie at $u < 0$ and are never met for $\tau > 0$. $\square$

The step $|\int g| \leq \int |g|$ is lossless here: in the coordinate (2) the integrand is a real magnitude times a deterministic phase, so no arithmetic cancellation is discarded, and the C, D -dependence factors out as a pure power. There is no oscillation estimate. The bound is numerically certified: $B_{1,0,1}(\Phi) = 0.686$ for a Gaussian Φ , and the certificate margin $\sup_x |A(x)| x^{\tau/2} / B = 0.914 \leq 1$ holds across $x \in [0.05, 2500]$ and $(\tau, m) \in \{0.5, 1, 2, 4\} \times \{0, 2\}$.

Theorem 2 (the sharp expansion, Part II Thm A.14). *With h_a, Φ as above, for every M ,*

$$I(C, D) := \int_{-1}^1 h_a(x) \Phi(C/\sqrt{1-x^2}) e(xD) dx = \sum_{m=0}^M \frac{e(\pm D) A_m^{\pm}(\mp C^2 D)}{(\mp 2\pi i D)^{m+1+a/2}} + R_M(C, D),$$

with $A_m^{\pm}$ as in (2) and $|R_M| \leq \mathcal{B}_{M, \tau_1} (C^2 D)^{-\tau_1/2} D^{-(M+1+a/2)}$, $\mathcal{B}_{M, \tau_1}$ independent of C, D .

Proof. Localization. Fix a smooth partition $1 = \chi_+ + \chi_0 + \chi_-$ with $\chi_{\pm}$ supported in $|x \mp 1| < \delta$ and χ_0 in $[-1 + \delta, 1 - \delta]$. On χ_0 the phase $x \mapsto x$ has nonvanishing derivative, so the non-stationary phase lemma [5, Thm. 7.7.1] (N -fold integration by parts) gives an $O_N(D^{-N})$ contribution with implied constant controlled by $\|\partial_x^N(\chi_0 h_a \Phi(C/\sqrt{1-x^2}))\|_{L^1}$. This is bounded uniformly in C : on $[-1 + \delta, 1 - \delta]$ the factor $\Phi(C/\sqrt{1-x^2})$ and its x -derivatives are bounded uniformly in C , since for Schwartz Φ and $c \geq \delta' > 0$ one has $\sup_{C>0} |C^k \Phi^{(k)}(Cc)| < \infty$.

Endpoint reduction. Near $x = 1$ substitute $x = 1 - t$, $t \in (0, \delta)$: then $h_a(1 - t) = t^{a/2} \varphi(t)$ with $\varphi(t) = (2 - t)^{a/2} h_1(1 - t) \in C^\infty[0, \delta]$; $\sqrt{1 - x^2} = \sqrt{t} \sqrt{2 - t}$; and $e(xD) = e(D) e(-tD)$, so $I_+ = e(D) \int_0^\delta \chi_+(1 - t) t^{a/2} \varphi(t) \Phi(C/(\sqrt{t} \sqrt{2 - t})) e(-tD) dt$.

Mellin representation and closed-form t -integral. For $\sigma > 0$ in the strip of holomorphy of $\tilde{\Phi}$, insert $\Phi(w) = \frac{1}{2\pi i} \int_{(\sigma)} \tilde{\Phi}(u) w^{-u} du$ with $w = C/(\sqrt{t} \sqrt{2 - t})$. The joint integrand is absolutely integrable on $\{\operatorname{Re} u = \sigma\} \times (0, \delta)$ — the t -integral converges absolutely and $\tilde{\Phi}(\sigma + iv)$ decays exponentially in v — so Fubini applies. Writing $\psi_u(t) = \varphi(t) (2 - t)^{u/2} \chi_+(1 - t) \in C^\infty$ and Taylor-expanding $\psi_u(t) = \sum_{m=0}^M b_m(u) t^m + R_M(u, t)$, the m -th t -integral is, by Watson’s lemma / the

Mellin evaluation of an oscillatory monomial [6, Ch. 3] (the smooth cutoff at $t = \delta$ costs only $O(D^{-\infty})$),

$$b_m(u) \int_0^\infty t^{m+\frac{a+u}{2}} e(-tD) dt = b_m(u) \Gamma(m+1+\frac{a+u}{2}) (2\pi i D)^{-(m+1+\frac{a+u}{2})} + O(D^{-\infty}).$$

The oscillation is evaluated in closed form, with no cancellation estimate. Since $C^{-u} D^{-u/2} = (C^2 D)^{-u/2}$, collecting the u -contour integral gives the m -th term $e(D)(2\pi i D)^{-(m+1+a/2)} A_m^+(C^2 D)$ with A_m^+ exactly (2): the coefficients $b_m(u)$, carrying the $(2-t)^{u/2}$ Taylor data, reproduce Altuğ's $c_m^\pm$.

Remainder. The Taylor remainder satisfies $|R_M(u, t)| \leq C_M(u) t^{M+1}$ with $C_M(u) = \sup_{[0, \delta]} |\partial_t^{M+1} \psi_u| / (M+1)!$ of at most polynomial growth in $\Im u$ (ψ_u smooth). Its contribution is again a transform of the form (2) with amplitude R_M ; taking its u -contour at $\operatorname{Re} u = \tau_1$ (any $\tau_1 > 0$), Lemma 1 bounds it by $\mathcal{B}_{M, \tau_1} (C^2 D)^{-\tau_1/2} D^{-(M+1+a/2)}$ with $\mathcal{B}_{M, \tau_1} < \infty$ independent of C, D . The endpoint $x = -1$ gives the $e(-D)$, A_m^- terms identically. $\square$

Remark 3 (on the exponents). The contour height $\tau_1 > 0$ is free, and the remainder decays like $(C^2 D)^{-\tau_1/2}$ because the factor $(C^2 D)^{-u/2}$ is evaluated at $\operatorname{Re} u = \tau_1$; Altuğ's stated $(C^2 D)^{-\tau_1}$ is the same statement in his normalization of the Mellin variable (u versus $u/2$). What the application uses is only that, for arbitrary $A > 0$, the remainder is $O_A((C^2 D)^{-A} D^{-(M+1+a/2)})$ uniformly in C, D — immediate from the freedom in τ_1 together with Lemma 1.

Proposition 4 (Part III Prop. 5.2). $|J_{l,f}(\xi, \nu, X)| \ll X^2 \nu^{-M} (lf^2)^{-N} ((lf^2/\sqrt{X})^{N-M+3} + \xi^M)$ with implied constant independent of (ξ, ν, l, f, X) .

Proof. Write $J_{l,f}(\xi, \nu, X) = \int G(y/X) y \Psi(y) e(-y\nu/2lf^2) dy$ with $\Psi(y) = I_{l,f}(\xi, y)$. Since G is compactly supported, integrating by parts M times against the phase gives the exact identity $J = (lf^2/(-\pi i \nu))^M \int \partial_y^M (G(y/X) y \Psi(y)) e(-y\nu/2lf^2) dy$, hence

$$|J_{l,f}(\xi, \nu, X)| \leq \left(\frac{lf^2}{\pi \nu} \right)^M \left\| \partial_y^M (G(y/X) y \Psi(y)) \right\|_{L^1(y)}.$$

By the Leibniz rule $\partial_y^M (G(y/X) y \Psi) = \sum_{i+j+k=M} \binom{M}{i,j,k} X^{-i} G^{(i)}(y/X) (y)^{(j)} \partial_y^k \Psi$, where $(y)^{(j)} = y, 1, 0$ for $j = 0, 1, \geq 2$. The cutoff $G^{(i)}(y/X)$ is supported on $y \in [\frac{X}{4}, \frac{5X}{4}]$ (length $\asymp X$) and contributes $\|G^{(i)}\|_\infty$, and the factor $(y)^{(0)} = y \asymp X$ supplies a second power of X , giving the X^2 . Each y -derivative of $\Psi(y) = \int \theta_\infty(x) [F + H](\arg) e(-x\xi\sqrt{4y}/4lf^2) dx$ falls on either the phase — producing a factor $O(\xi/lf^2\sqrt{y})$, the source of the ξ^M branch — or on the profile through $\arg = lf^2/(\sqrt{4y}\sqrt{1-x^2})$, producing a factor $O(lf^2/y)$, the source of the $(lf^2/\sqrt{X})$ branch. Each resulting x -integral is a transform of the form (2) (with $C \asymp \xi\sqrt{y}/lf^2$ and D the local scale), bounded by Lemma 1, or at the sharp order by the expansion of Theorem 2. Collecting the X -, lf^2 -, and ξ -powers reproduces the two branches $(lf^2/\sqrt{X})^{N-M+3}$ and ξ^M ; the implied constant is a finite sum of the integrals $B_{\tau,m,a}(\Phi)$ (Lemma 1) and the norms $\|G^{(i)}\|_\infty$, none depending on (ξ, ν, l, f, X) . $\square$

Every uniform constant in §2 is one application of Lemma 1; the content of Altuğ's Appendix A, from the exact-gauge standpoint, is that single magnitude estimate.

3 The oscillation is the cutoff's comb

Why did Prop. 4 look hard? The oscillation the estimate must survive is deterministic. Scanning $|J_{l,f}(\xi, \nu, X)|$ in the transform frequency ν over the cutoff support $y \in [\frac{X}{4}, \frac{5X}{4}]$ reveals a comb of

near-zeros with spacing, to measurement, $\Delta\nu = 4lf^2/(\kappa X)$ with $\kappa \approx 0.94$ the support fraction, matching the two-edge beat of G : the dispersion law $\Delta\nu \propto (lf^2)^{1.00}(X/8)^{-0.98}(1+\xi)^{0.02}$ fits at $R^2 \approx 1$ (Appendix A) — the interference of the two edges of the fixed cutoff, switched on once the ξ -phase drives the x -integral into edge dominance. At $\xi = 0$ the x -integrand is unimodal (prominence ratio $P_2/P_1 = 0$) and the comb is absent. The apparent difficulty is a chart feature of the fixed cutoff—exactly what the exact gauge of §2 removes.

4 The productivity boundary

The symmetric-power kernel is modulated by $U_r(\cos\theta) = \sum_{j=0}^r e^{i(r-2j)\theta}$, carrying a zero-frequency component iff r is even. At $\xi = 0$ the x -integral reads this component (the detecting residue); at $\xi \neq 0$ it is the removable comb of §3, bounded by Prop. 4 (with $|U_r| \leq r+1$) and hence $o(X)$. The two regions are disjoint in ξ : the detecting residue sits at $\xi = 0$, where the comb is absent; the comb sits at $\xi \neq 0$, where it is removable. So the residue isolates for every r . On the dual integral (level $p^k = 25$) the zero-frequency value against the $\xi \neq 0$ floor is

$$r = 1 : 0 \text{ (exact)}, \quad r = 2 : 55.9, \quad r = 3 : 1.4 \times 10^{-16}, \quad r = 4 : 11.4,$$

nonzero iff r even, clear of the floor by an order of magnitude. The degradation with r is the growth of the character tail mass (0.042, 0.092, 0.135, 0.156, 0.256 for $r = 0, \dots, 4$, fit ceiling $\times \sqrt{\#\text{harmonics}}$ at $R^2 = 0.977$)—gradual attrition, not a wall. Sarnak’s productivity boundary [3, fn. 5] records the growth of a removable floor, not a loss of the residue.

5 Toward functoriality: the product law and the transfer factor

The three preceding sections close the analytic core — the uniformity (§2), the emergent clock (§3), and the productivity boundary. The remainder of the paper develops the two mechanisms toward symmetric-power functoriality, beginning with the product law, which supplies the transfer factor.

The derivative statistic that random-matrix theory reads at an eigenvalue is the reopening rate an arithmetic fiber reads at a zero, by (1): both are $\prod_{r_j \neq \mu} |\mu - r_j|$. This is an identity, not a coincidence, and it is the mechanism the ensemble leaves implicit. The consequence here is structural: the same distance product is a *Weyl denominator*, and Weyl denominators are the archimedean transfer factors of §7. Self-interference is one object appearing as a zero statistic, a reopening rate, and a transfer factor.

6 The carrier/fiber rescaling: a pole becomes a closed cell

Place the integers on a carrier scaled by $\pi/3$, i.e. attach the cell phase $e^{in\pi/3}$ (the μ_6/ζ_6 step). The pole of ζ at $s = 1$ lives in the divergence of $\sum 1/n$; on the carrier it closes,

$$\sum_{n \geq 1} \frac{e^{in\pi/3}}{n} = -\log(1 - e^{i\pi/3}) = \frac{i\pi}{3},$$

a finite μ_6 value: the divergence has become a closed-cell reading. For a Dirichlet character χ of conductor q , the appropriate fiber scale (trivial $\rightarrow \pi/6$, $\chi_3 \rightarrow \pi/3$, $\chi_4 \rightarrow \pi/2$) makes carrier and fiber compatible harmonics; the principal character, carrying the pole, is handled by the eta

regulator,

$$\sum_{n \geq 1} \frac{(-1)^{n-1} e^{in\pi/3}}{n} = \log(1 + e^{i\pi/3}) = \frac{1}{2} \ln 3 + \frac{i\pi}{6},$$

landing on the trivial character's own cell $(\pi/6, \mu_{12})$, pole-free, with the partial-sum phase converging (no drift). In this frame the $S(t)$ oscillation of the readout is removed—the value is a fixed harmonic, not a wandering phase—so what was a residue to be identified is a zero-frequency cell value, read as the DC-residue $(s-c)F'/F \rightarrow r$ of a logarithmic derivative. This is the reading used in §8.

7 Symmetric-power transfer: arithmetic transparency

For Sym^r transfer $\text{GL}(2) \rightarrow \text{GL}(r+1)$ ($r = 2m$ even), the transfer sends an elliptic class of Satake angle θ to the class of angles $\{2k\theta : |k| \leq m\} \cup \{0\}$ —i.e. Sym^r of the Frobenius block, with the fixed eigenvalue $1 = \alpha\beta$ the det-one of the block. Two features make the archimedean transfer factor explicit and the arithmetic transparent.

Arithmetic transparency. The transferred characteristic polynomial is reducible ($x-1$ divides it), and each quadratic factor $x^2 - 2\cos(2k\theta)x + 1$ has discriminant $-4\sin^2(2k\theta)$. The clock identity $\sin(2k\theta) = \sin\theta U_{2k-1}(\cos\theta)$ (with $\sin^2\theta = |D|/4n$, $D = t^2 - 4n$) gives

$$\frac{\text{disc}_k}{D} = \frac{U_{2k-1}(\cos\theta)^2}{4n} \in \mathbb{Q}^{\times 2},$$

a rational square—so every factor lies in the *same* field $\mathbb{Q}(\sqrt{D})$ as the original class, the arithmetic L -value is identical on both sides, and it cancels in the transfer factor. (Because symmetric powers transfer to *reducible* classes, this avoids the order-zeta conjecture of Deng–Espinosa, which concerns irreducible cubic orders.)

Transfer factor as a distance product. What remains is the ratio of Weyl denominators — distance products by (1). For $r = 2$,

$$\frac{\Delta_3}{\Delta_2} = \frac{|1 - e^{2i\theta}|^2 |e^{2i\theta} - e^{-2i\theta}|}{|e^{i\theta} - e^{-i\theta}|} = 4|\sin\theta||\sin 2\theta| = 8\sin^2\theta|\cos\theta| = \frac{|D||t|}{n^{3/2}},$$

a monomial in the class data; its winding phase, gauged by the $\pi/3$ (μ_6) step, is a real harmonic (§6). Thus $\Delta(\theta) = U_r(\cos\theta) \cdot (\Delta_{r+1}/\Delta_2) \cdot (\text{radial } \sqrt{p}\text{-scaling})$, the Weyl character and distance product being product-law objects and the phase an exact-gauge object.

8 Detection: the self-twist identity closes

We isolate one computable instance of the detection step and close it on a genuine transfer. For a holomorphic newform f with a_p its Frobenius trace and $\lambda(p) = a_p/p^{(k-1)/2}$, the symmetric square carries the middle eigenvalue $\alpha\beta = 1$; the plain census $\sum_p (\lambda(p)^2 - 1)(\log p)/p$ has mean zero and is blind. The correct object is the *self-twist* Rankin–Selberg L -function $L(s, f \times f \otimes \chi_K)$ for the quadratic character χ_K of an imaginary quadratic field K : it has a pole at $s = 1$ iff f has complex multiplication by K (dihedral), the classical criterion of Gelbart–Jacquet and Rankin–Selberg. Its pole order is the transferred count, read as the zero-frequency residue of the logarithmic derivative:

$$(\text{LEFT, analytic}) \quad A_{\text{tw}}(X) = \sum_{p \leq X} \lambda(p)^2 \chi_K(p) \frac{\log p}{p}, \quad \frac{d A_{\text{tw}}}{d \log X} \longrightarrow \text{ord}_{s=1} L(s, f \times f \otimes \chi_K).$$

The geometric side is the self-twist relation on Frobenius, $f \otimes \chi_K \cong f$, equivalently $a_p = 0$ at every inert prime:

$$(\text{RIGHT, geometric}) \quad \#\{p \leq X \text{ inert} : a_p = 0\} / \#\{p \leq X \text{ inert}\} \longrightarrow 1_{\{f \text{ has CM by } K\}}.$$

We evaluate both on two weight-two forms with $K = \mathbb{Q}(\sqrt{-3})$ (the Eisenstein field, whose $\chi_K(p) = +1, -1$ for $p \equiv 1, 2 \pmod{3}$): the CM curve $y^2 = x^3 + 1$ and the non-CM curve $y^2 = x^3 - 16x + 16$ (conductor 37), with a_p from point counting.

	LEFT: slope $dA_{\text{tw}}/d \log X$	RIGHT: inert $a_p = 0$ fraction	count
CM $\sqrt{-3}$ ($x^3 + 1$)	0.997	725/726 = 0.9986	1
non-CM ($x^3 - 16x + 16$)	0.037	6/726 = 0.0083	0

For the CM form the analytic slope is 0.97–0.99 across every interval [1000, 4000], [4000, 10000], [10000, 20000] (pole order 1), and the geometric self-twist holds at every good inert prime (the single exception 725/726 is $p = 2$, the bad-reduction prime of conductor $36 = 2^2 \cdot 3^2$). For the non-CM form both sides are 0. Thus

$$\text{ord}_{s=1} L(s, f \times f \otimes \chi_K) = 1_{\{f \otimes \chi_K \cong f\}} = \text{transferred count}$$

closes—1 on a genuine transfer, 0 on a genuine non-transfer—read through the logarithmic- derivative DC-residue on the $S(t)$ -free carrier of §6, with the emergent-clock lesson explaining why the plain census cancels (inert primes carry $\lambda^2 = 0$, so the self-twist census has no cancelling floor).

Remark 5 (what this is). The mathematics detected here—that the symmetric-square/self-twist pole marks CM forms—is classical (Gelbart–Jacquet, Rankin–Selberg). What the section contributes is the closure of the detection *identity* through the two mechanisms: the analytic count as a product-law/log- derivative residue, the geometric count as the self-twist relation, matched on a genuine transfer.

8.1 A non-abelian instance: which Frobenius data transfers

The self-twist detection of §8 runs on a dihedral (abelian, C_2) form, where the Galois image is small and the arithmetic reduces to a single quadratic character. For a genuinely non-abelian form the Frobenius datum splits into two independent bits—one a character cannot see—and the transfer treats them differently. We illustrate this on the exotic tetrahedral (“ A_4 ”) weight-one newform h of level 124, whose L -coefficients are a class function of Frob_p in the binary tetrahedral group $2T = \text{SL}(2, 3)$, given by the closed formula $a_p = \zeta_{12}^{k(p)}$ with

$$k(p) = \frac{1}{2}(\chi_k(p) + 4\varepsilon(p)) + (0 \text{ if } (3/p) = +1, \text{ else } 6) \pmod{12},$$

$a_p = 0$ when $\varepsilon(p) = 0$. Here the exponent carries two independent Galois bits:

value bit $\varepsilon(p) \in \{0, \pm 1\}$ (the cubic residue, the abelian $A_4^{\text{ab}} = C_3$ datum), *sign bit* $(3/p)$ (the $\text{SL}(2, 3) \rightarrow A_4$ cen

the double-cover datum read from the order of Frob_p ; the sign bit resolves k versus $k + 6$ and is invisible to any abelian character.

The symmetric square sends $a_p \mapsto a_p^2 - \chi(p)$, and $a_p^2 = \zeta_{12}^{2k}$ is unchanged by $k \mapsto k + 6$. Hence:

Sym ² transfer to GL(3)	
sign bit (3/p) (double cover, SL(2, 3))	<i>forgotten</i> (trace invariant under $k \leftrightarrow k+6$)
value bit ε (cubic residue, projective A_4)	<i>retained</i> : $\varepsilon = \pm 1 \Rightarrow \pm\sqrt{3}i$, $\varepsilon = 0 \Rightarrow \pm 1$

So the transfer factors through the projective image A_4 : the value bit passes to the GL(3) Satake datum, the sign bit is the GL(2)-specific double-cover fiber the transfer collapses. The order trick—sign from the element order, value from the residue symbol—thus reads *exactly which part of the Frobenius datum transfers*, and the split is non-trivial precisely because A_4 is non-abelian (for the dihedral case of §8 the two bits coincide). This identifies the transferred datum; the automorphy of the resulting GL(3) object (the three-dimensional standard representation of A_4) is the classical Artin/Langlands–Tunnell case.

8.2 The detected count is a Mordell–Weil rank difference

The transferred object of §8.1 is, up to the nebentype twist, the three-dimensional adjoint $\text{Ad}_g = \text{End}^0(V_g) \cong \text{Sym}^2 V_g \otimes (\det V_g)^{-1}$ of the A_4 form g . At this exotic locus the zero-frequency count read on the analytic side coincides with a genuine arithmetic rank, and the coincidence is an exact identity of group theory.

The quartic field M cut out by g carries the permutation action of A_4 on four points, whose character is $\chi_{\text{perm}} = (4, 0, 1, 1)$ on the classes $(e, (2, 2), (3)_a, (3)_b)$. Against the character table, $\langle \chi_{\text{perm}}, \mathbf{1} \rangle = 1$ and $\langle \chi_{\text{perm}}, \chi_{\text{std}} \rangle = 1$, so

$$\chi_{\text{perm}} = \mathbf{1} \oplus \text{Ad}_g, \quad \text{Ad}_g = \chi_{\text{std}} \text{ (the 3-dimensional irreducible).}$$

Functoriality of Mordell–Weil under this decomposition gives $E(M) \otimes \mathbb{Q} = (E(\mathbb{Q}) \otimes \mathbb{Q}) \oplus E(\cdot)_{\text{Ad}_g}$ and hence the *exact* Artin/permutation identity

$$r(E, \text{Ad}_g) = \text{rank } E(M) - \text{rank } E(\mathbb{Q}), \quad (3)$$

proven as group theory (no analytic or p -adic input). The left side is the order of vanishing at $s = 1$ of $L(E, \text{Ad}_g, s)$ —the zero-frequency residue of the logarithmic derivative, the same statistic §8 read as the transferred count. The right side is a difference of classical Mordell–Weil ranks, the source living “up the tower” at M and traced down to $\mathbb{Q}$. Thus the count is a rank difference: one statistic, read once on the automorphic side and once on the arithmetic side.

On the two Darmon–Lauder–Rotger exotic A_4 curves this is pinned on both sides:

	rank $E(\mathbb{Q})$	rank $E(M)$	$r(E, \text{Ad}_g) = \text{rank } E(M) - \text{rank } E(\mathbb{Q})$
$26b \ (y^2 + xy + y = x^3 - x^2 - 3x + 3)$	0	2	2
$52b \ (y^2 = x^3 + x - 10)$	0	2	2

with $M = \mathbb{Q}(w)$, $w^4 + 7w^2 - 2w + 14 = 0$ the tetrahedral field of §8.1, rank-two generators of $E(M)$ exhibited by descent. This is the “one mechanism, two locations” statement made theorem-anchored: at the exotic locus both sides of (3) are determined, the identity between them is proven group theory, and the shared statistic is the DC-residue that reads the functorial pole in §8 and the analytic rank in the Beyond-Endoscopy census alike.

Remark 6 (tiering). The rank identity (3) is proven (Artin formalism). The automorphy of Ad_g on GL(3)—which makes $L(E, \text{Ad}_g, s)$ an honest automorphic L -function so that its pole order is

a spectral quantity—is the classical Langlands–Tunnell case for A_4 . The p -adic *regulator* side (the explicit point coordinates realizing the rank) is the Darmon–Lauder–Rotger Elliptic Stark conjecture, verified to 20 digits on $26b/52b$ but with the points found by descent first; we use only the proven rank identity and the classical automorphy here.

9 A candidate automorphic object from the primary construct

The transfer of §§7–8.2 requires an automorphic representation on $\mathrm{GL}(r+1)$ whose existence realizes the functorial lift. In the primary (three-dimensional) representation we exhibit a concrete *candidate object* for it: the *carrier* — a double-sided helix of Frobenius similitudes — carrying a *fiber* (the phasor readout), together with its *warp* (the twist family of §7). The classical L -function is the shadow of this candidate under the projection that sends the exponential state space e^y down by a Möbius/Cayley map and then log to the ordinate, at the midpoint and without drift. What makes it a candidate is that the three properties a converse theorem asks of an L -function — functional equation, analytic continuation, and boundedness/completeness — are each a *proven* statement about it, and each is already formalized.

Functional equation = the helix–conjugate link. The anti-helix is the complex *conjugate* of the helix, not its mirror: the carrier’s transverse block is $\mathrm{diag}(z, \bar{z})$ with z the Frobenius spin, and

$$z \cdot \bar{z} = |z|^2 = 1$$

(`spin_mul_conj`), so the block has determinant one and is unitary (`frobeniusBlock_det_one`, `frobeniusBlock_unitary`). This $z\bar{z} = 1$ *is* the functional equation — the pairing of s with $1-s$ through the conjugate leg. A unit-modulus warp $A(n) = e^{i\theta(n)}$ sends $z \mapsto Az$, $\bar{z} \mapsto \bar{A}\bar{z}$, hence $z\bar{z} \mapsto |A|^2 = 1$: the det-one link is preserved for the *entire* warp family, so every twist carries a functional equation, structurally and without estimate. (A complex twist is *self*-dual only when it equals its own conjugate; otherwise the equation pairs the sideband χ_j with $\chi_{-j} = \bar{\chi}_j$, as the conjugate leg demands.)

Continuation and the completed function = Tate’s thesis. The fiber’s value is the phasor representation $L(\chi, s) = \sum_n \chi(n) n^{-s}$ `mellinSpin(y, n)` with unit-modulus, multiplicative readout spin. Its meromorphic continuation and the completed function $\Lambda(s) = \gamma(s)L(s)$, carrying the archimedean γ -factor, are supplied by the adelic zeta integral of Tate’s thesis [7] for the $\mathrm{GL}(1)$ constituents and their twists (and by Godement–Jacquet [8] for the standard L -function of an automorphic representation on $\mathrm{GL}(n)$). In that formalism the functional equation is assembled from the *local* functional equations — one per place, each the determinant-one conjugate pairing $z\bar{z} = 1$ above — by Poisson summation on $\mathbb{A}/\mathbb{Q}$. So the completed function is precisely the Tate assembly of the per-place det-one links: continuation and functional equation follow the standard adelic route, and the primary representation’s contribution is that the local inputs it consumes are exactly the carrier’s determinant-one blocks.

Boundedness/completeness = three-dimensional exhaustion. The remaining input — that the object has no spectral mass unaccounted for, uniformly over the family — is, in the primary representation, the statement that every fiber is *exhaustive* over its event space. The event space of the three-dimensional object is the height ray, which is the weld; there is nowhere

off-weld for a zero event to occur, so every zero event is weld-fixed and is a source. This is proven unconditionally, for *every* fiber with no hypotheses:

$$\text{threeD_exhaustive } (E : \mathbb{C} \rightarrow \mathbb{C}) : \quad \forall y \in \mathbb{R}, E(y) = 0 \Rightarrow \text{IsSource } E(y).$$

The proof is short for an ontological reason: the event space of the primary object is the height ray itself, and the strip is a one-dimensional projection device with no primary counterpart, so there is nowhere for an unaccounted mode to hide. This is the primary-representation form of the family-wide boundedness the converse theorem requires.

The candidate, assembled. On the primary construct the object exists, its functional equation holds across the whole warp family (det one, `frobeniusBlock_det_one`), its continuation is the readout, and its completeness is exhaustion (`threeD_exhaustive`) — each a proven, formalized statement. The carrier, fiber, and warp *are* the candidate automorphic object, and its three converse-theorem inputs are discharged in the primary representation. The bridge from the candidate to the classical statement that $\text{Sym}^r \pi$ is automorphic on $\text{GL}(r+1)$ is the converse theorem, whose three hypotheses are exactly the three proven properties above; supplying them is what the candidate does. This is the object we use to realize the functorial lift of §§7–8.2. All four inputs — the functional equation surviving every warp (angle), the phasor/Abel readout (continuation), three-dimensional exhaustion (height), and the loss-ledger reconstruction bijection (recovery) — are bundled and machine-checked in a single theorem (`AutomorphicCandidate.candidate_wellformed`), with the standard axiom footprint.

Recovery: the loss-ledger round trip. The projection to the shadow is lossy but bookkept exactly, and the descent is dimensional: from the three-dimensional object the *radial* channel is dropped ($3\text{D} \rightarrow 2\text{D}$) and then the *angular* channel ($2\text{D} \rightarrow 1\text{D}$), while the height passes by the log/exp map between the exponential state space e^y and the ordinate y . Recording the two dropped channels in a ledger, the map $\text{fiber} \mapsto (\text{ordinate}; \text{radius}, \text{angle})$ is a *bijection* (`record_bijective`, `reconstruct_record`: an exact two-sided inverse, unconditional), provided the projection is taken at the midpoint, without drift. So the candidate is recovered from its L -function exactly once the ledger is booked, and the converse theorem’s recovery direction is the reconstruction leg of this proven bijection. The three ledger channels are precisely the three inputs: the *angle* is the functional-equation link $z\bar{z} = 1$, the *height* is exhaustion, and the *radius* is the area-law scaling $\sqrt{p}$ of the Frobenius similitude.

Remark 7 (a separate matter). The same candidate object carries a distinct and independent question — the location of the zeros of its one-dimensional shadow, i.e. the Riemann/generalized Riemann hypothesis for that shadow. That question is the subject of a completely separate development, is addressed nowhere in the present paper, and is logically independent of everything above: the candidate’s existence, its functional equation, and its exhaustion are established without any reference to where its zeros lie. We make no claim, in any direction, about RH or GRH here.

10 Status and active research

We summarize the standing of each result at its own strength.

- **Theorem** (§2). The archimedean uniformity—Altuğ’s “central issue”—follows from the single magnitude bound Lemma 1, with Theorem 2 and Proposition 4 obtained by standard

reductions in which every uniform constant is one application of the lemma. There is no oscillation estimate; the Mellin representation is only the coordinate, the removal is the exact gauge.

- **Structural** (§§5,6,7). The product law identifies the GUE derivative statistic, the reopening rate, and the Weyl-denominator transfer factor as one object. The carrier/fiber rescaling turns a pole into a closed-cell reading and removes $S(t)$ from the readout. The symmetric-power transfer has transparent arithmetic (same field $\mathbb{Q}(\sqrt{D})$, order-zeta conjecture avoided) and an explicit transfer factor.
- **Measured** (§§3,4,8). The emergent comb is the cutoff's edge kinematics ($R^2 \approx 1$); the productivity boundary is gradual attrition ($R^2 = 0.977$), not a wall; the symmetric-square detection identity closes on a genuine transfer (count 1) and a genuine non-transfer (count 0).
- **Formalized** (§9). The candidate automorphic object's four converse-theorem inputs — functional equation (surviving every warp), continuation (Tate's completed function / the phasor readout), three-dimensional exhaustion, and loss-ledger recovery — are machine-checked in Lean (Mathlib), theorem `candidate_wellformed`, with the standard axiom footprint and no `sorry/axiom`, built in-tree.

Active research. A full symmetric-power functoriality theorem—the statement that $\text{Sym}^r \pi$ is automorphic on $\text{GL}(r+1)$ —is the subject of active research. The present paper isolates its ingredients and settles several: the archimedean uniformity is a theorem (§2); the fundamental-lemma arithmetic is transparent for the whole even family (§7); the detection identity closes on a worked transfer (§8); and the candidate object's converse-theorem inputs are discharged in the primary representation (§9). The remaining ingredient is the classical converse theorem itself: the identification of the candidate's L -function with a term of the $\text{GL}(r+1)$ spectrum (the symmetric-power analogue of Altuğ's Theorem 1.1), together with the uniform analytic control of the twisted family $L(s, \text{Sym}^r \pi \times \sigma)$ over σ — whose functional equations are the det-one links of §9, leaving the boundedness aggregate over the family as the object currently under study. The exact gauge governs the archimedean part; the product law and the carrier/fiber rescaling reduce the residue reading to cell arithmetic.

A Reproducibility

Every measured statement is produced by a self-contained script (no L -function library); the values quoted in the text are reproduced below with the script that computes each.

- **Lemma 1 (the gauge bound)**, `be_uniformity_certify.py`. For a Gaussian Φ , $B_{1,0,1}(\Phi) = 0.686$, and the certificate margin $M(x) = |A(x)| x^{\tau/2} / B$ satisfies $\sup_x M(x) = 0.914 \leq 1$, uniformly across $x = C^2 D \in [0.05, 2500]$ and $(\tau, m) \in \{0.5, 1, 2, 4\} \times \{0, 2\}$. The line integrand decays as $e^{-\pi|v|}$ ($v \rightarrow +\infty$) and $e^{-\pi|v|/2}$ ($v \rightarrow -\infty$), so $B_{\tau,m,a}(\Phi) < \infty$ for every $\tau > 0$.
- **Proposition 4 (the uniform constant)**, `be_prop52_certify.py`. The constant $L_M = \|\partial_y^M(G(y/X) y \Psi)\|_{L^1}$ fits a uniform monomial $L_M \sim C_M(lf^2)^{a_M}(1+\xi)^{b_M} X^{g_M}$ with $a_M = 0.41, 0.05, -0.05$ for $M = 1, 2, 3$ and $b_M \leq 0$. The lf^2 -exponent a_M grows more slowly than M ($da/dM \approx -0.23$), so each integration by parts nets ν^{-1} and the bound is summable — the standard-representation side of the productivity boundary. (This is the crude magnitude route; the sharp exponents of Proposition 4 follow from Theorem 2.)

- **The emergent clock (§3), `mb_beat.py`.** The comb dispersion law $\Delta\nu \propto (lf^2)^{1.00}(X/8)^{-0.98}(1+\xi)^{0.02}$ fits at $R^2 \approx 1$ — i.e. $\Delta\nu = 4lf^2/(\kappa X)$, $\kappa \approx 0.94$, essentially independent of ξ — identifying the comb as the cutoff’s two-edge beat rather than arithmetic content; the same run confirms that the comb vanishes at $\xi = 0$ (the integrand is unimodal there).
- **The productivity boundary (§4), `mb_uniform.py`.** The Weyl-character tail law $\sim (r+1)\sqrt{\#\text{harmonics}}$ fits the measured tail mass 0.042, 0.092, 0.135, 0.156, 0.256 ($r = 0, \dots, 4$) at $R^2 = 0.977$, well above the scrambled baseline 0.237; the $\xi = 0$ detecting residue against the $\xi \neq 0$ floor is 0, 55.9, 1.4×10^{-16} , 11.4 for $r = 1, 2, 3, 4$.
- **The candidate object (§9).** The four converse-theorem inputs are formalized and machine-checked in Lean (Mathlib): `RequestProject/AutomorphicCandidate.lean`, theorem `candidate_wellformed` axiom footprint `{propext, Classical.choice, Quot.sound}`, no `sorry/axiom`; it builds in-tree against the full library.

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